

Kinematic Analysis of Theo Jansen Mechanism

Project Part B Report

Course: IE410 – Introduction to Robotics

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B.Tech ICT

Semester 4

Winter 2026

Abstract

This project presents the kinematic analysis and trajectory generation of a modified Theo Jansen walking mechanism using MATLAB simulation. The Theo Jansen mechanism is a single degree-of-freedom linkage system capable of generating gait-like trajectories resembling human walking motion. The project focused on implementing the linkage kinematics, generating end-effector trajectories, analyzing gait characteristics, and studying the effect of varying important link lengths on the generated walking trajectory.

The mechanism was modeled using geometric constraints and circle-circle intersection methods for solving linkage joint positions. Simulations were performed over a complete crank rotation cycle to generate smooth gait trajectories. The generated trajectories were compared qualitatively with reference gait patterns from literature.

The project also investigated the influence of variable link lengths on stride length and gait height. Animation and trajectory visualization confirmed the correctness and stability of the implemented mechanism.

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Chapter 1

Introduction

Walking mechanisms are important in robotics, rehabilitation systems, locomotion platforms, and bio-inspired engineering applications. Among various walking linkage systems, the Theo Jansen mechanism is widely recognized for its ability to generate natural walking trajectories using only a single degree of freedom.

The Theo Jansen mechanism consists of interconnected rigid links arranged to generate gait-like motion. Unlike conventional robotic walking systems requiring multiple actuators and complex control algorithms, the Theo Jansen mechanism produces gait-like motion using purely mechanical linkage constraints.

This project focuses on the simulation and kinematic analysis of a modified Theo Jansen mechanism using MATLAB.

Chapter 2

Objectives

- To understand the kinematics of the Theo Jansen walking mechanism.
- To implement the linkage system using MATLAB simulation.
- To generate stable gait-like trajectories using a single rotational input.
- To visualize linkage motion and end-effector trajectories.
- To analyze the effect of varying link lengths on gait characteristics.
- To compare simulated trajectories with reference gait patterns.

Chapter 3

Literature Background

The Theo Jansen mechanism is a well-known linkage system inspired by biological locomotion. It was originally developed by Theo Jansen for generating walking motion using interconnected rigid links.

Recent research has explored the use of modified Jansen mechanisms in rehabilitation systems and gait training applications.

Chapter 4

Theo Jansen Mechanism

The Theo Jansen mechanism consists of multiple rigid links connected through revolute joints to generate gait-like motion using a single rotational input.

4.1 Mechanism Structure

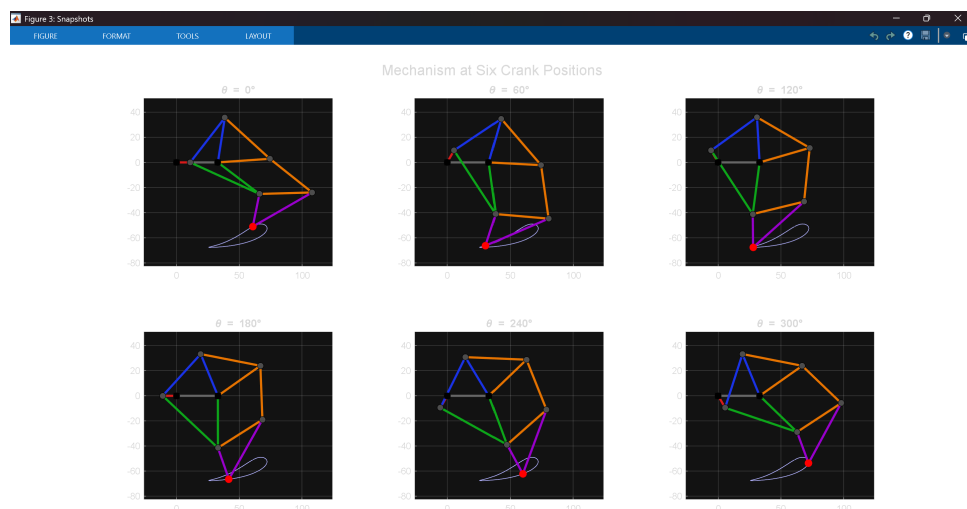


Figure 4.1: Theo Jansen mechanism structure at different crank positions

The mechanism contains interconnected rigid links that produce a periodic foot trajectory during crank rotation.

Chapter 5

Kinematic Modeling

The linkage positions were computed using geometric constraint equations and circle-circle intersection methods.

The rotational crank angle is defined as:

$$\theta = \omega t$$

The endpoint position of a rotating crank is:

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

The remaining joints were solved using circle-circle intersection equations:

$$(x - x_1)^2 + (y - y_1)^2 = r_1^2$$

$$(x - x_2)^2 + (y - y_2)^2 = r_2^2$$

Chapter 6

MATLAB Simulation Methodology

The mechanism was implemented using MATLAB numerical simulation.

The simulation procedure consisted of:

1. Define link lengths and fixed joints
2. Generate crank rotation over full cycle
3. Compute crank endpoint positions
4. Solve remaining linkage joints
5. Store end-effector coordinates
6. Generate gait trajectory plots
7. Animate mechanism motion

Chapter 7

Trajectory Generation

The generated trajectory resembles a gait-like walking pattern with stance and swing phases.

7.1 Generated Foot Trajectory

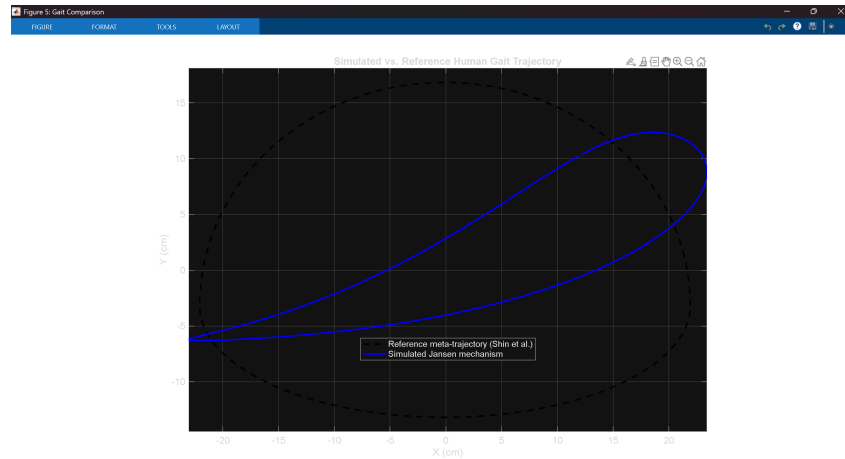


Figure 7.1: Generated foot trajectory

The generated trajectory qualitatively resembled a human gait-like curve with a relatively flat stance phase and elevated swing phase.

7.2 Gait Comparison

The generated gait trajectory was qualitatively compared with a reference gait-like trajectory from literature.

Minor deviations from the reference trajectory were observed due to simplified parameter selection and absence of trajectory optimization.

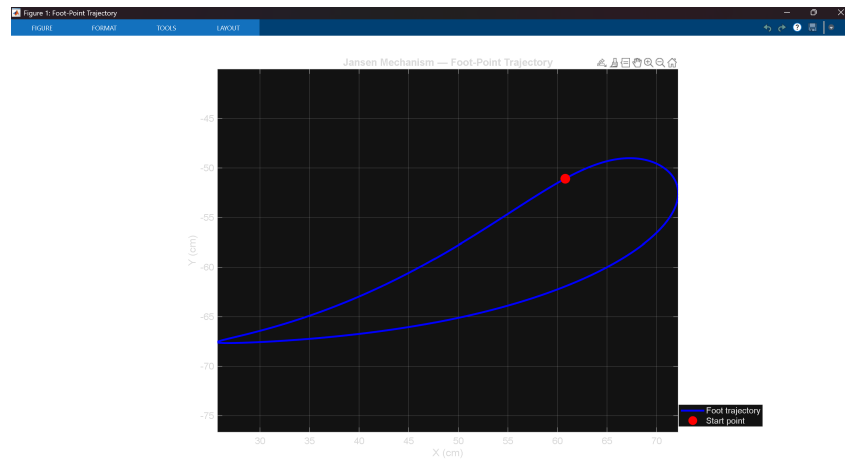


Figure 7.2: Qualitative comparison between simulated trajectory and reference gait curve

7.3 Gait Cycle Analysis

The variation of end-effector coordinates over the gait cycle was analyzed.

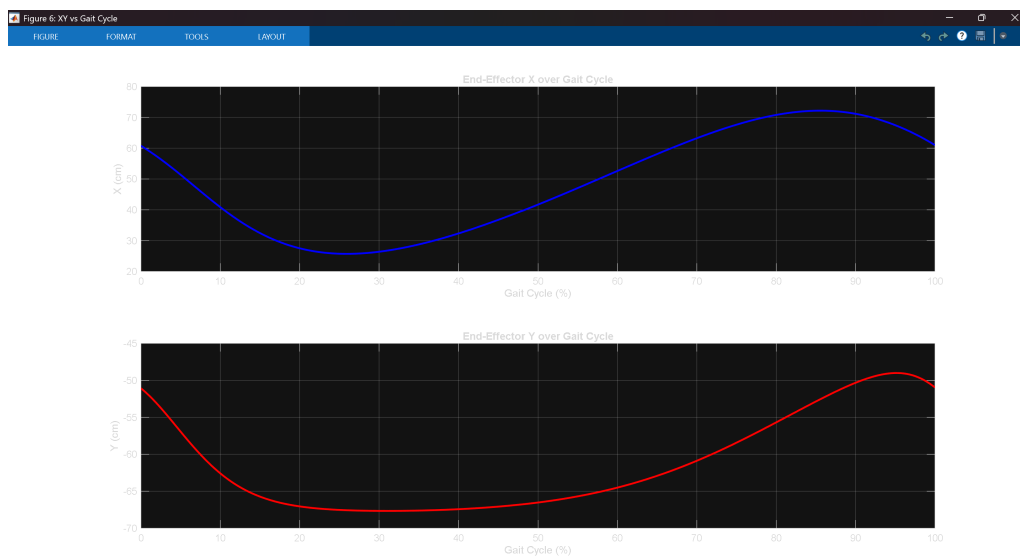


Figure 7.3: Variation of end-effector X and Y coordinates over gait cycle

The plots demonstrate smooth periodic variation in both horizontal and vertical directions.

Chapter 8

Link Length Variation Analysis

The effect of varying important linkage dimensions on the generated gait trajectory was analyzed.

Variation in link length L_4 primarily affected the horizontal stride length, while variation in link length L_8 significantly influenced the vertical gait height and swing elevation.

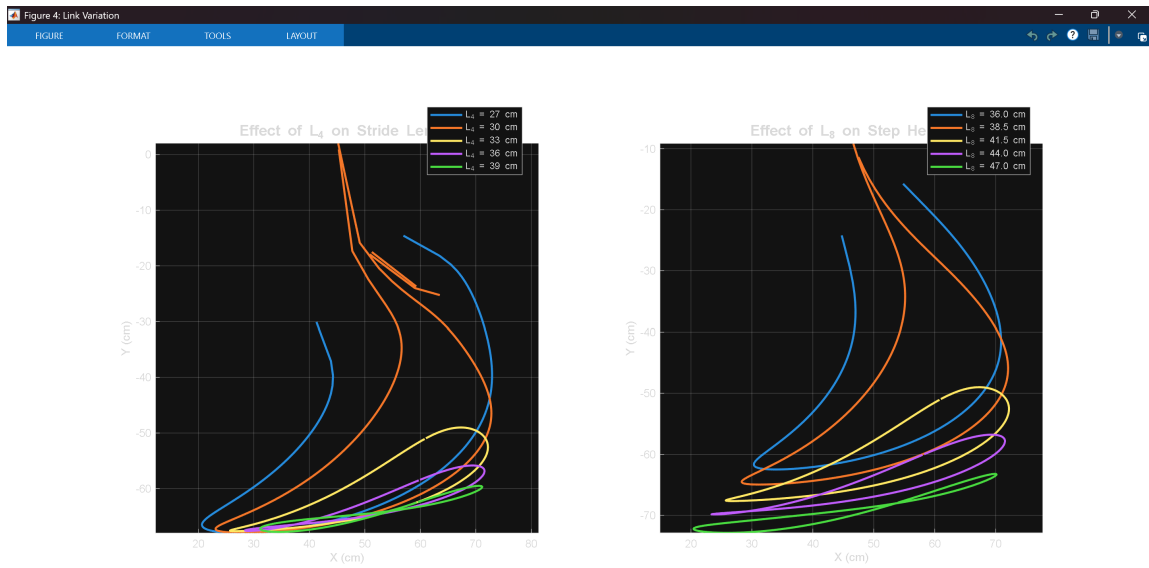


Figure 8.1: Effect of varying link lengths on gait trajectory characteristics

The simulation results demonstrated that gait characteristics can be significantly modified by adjusting selected linkage dimensions.

Chapter 9

Animation and Visualization

Mechanism animation was implemented to verify the correctness and continuity of the linkage motion.

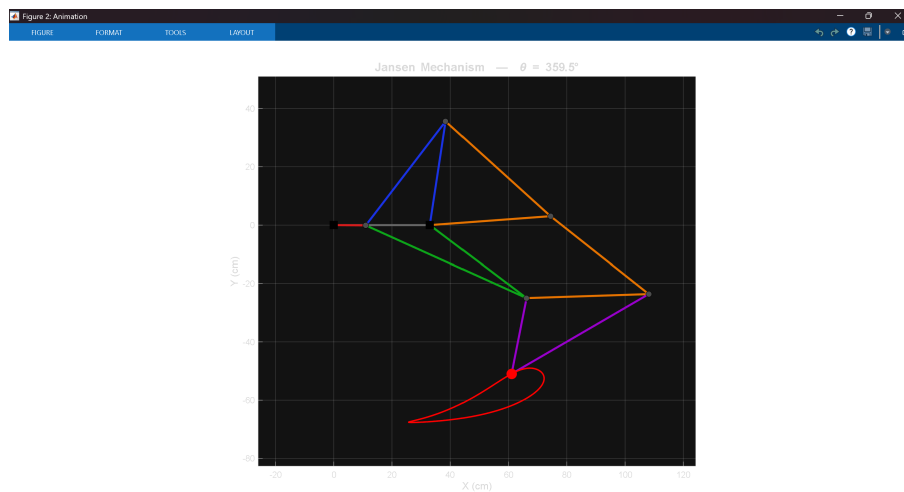


Figure 9.1: Animation snapshot of Theo Jansen mechanism

The animation confirmed stable linkage connectivity and smooth periodic motion.

Chapter 10

Results and Discussion

The MATLAB implementation successfully generated stable gait-like trajectories using the Theo Jansen linkage mechanism.

The generated trajectories exhibited:

- Stable periodic motion
- Smooth trajectory continuity
- Flat stance phase
- Elevated swing phase
- Continuous gait generation

The mechanism remained kinematically stable throughout the simulation.

Chapter 11

Challenges Faced

Several implementation challenges were encountered:

- Determining correct linkage topology
- Handling multiple circle intersection branches
- Maintaining stable linkage configurations
- Preventing disconnected motion
- Selecting meaningful link dimensions
- Matching gait trajectories with literature

Chapter 12

Conclusion

This project successfully implemented and analyzed a modified Theo Jansen walking mechanism using MATLAB simulation.

The mechanism generated stable gait-like trajectories using only a single rotational input. Link variation analysis demonstrated the influence of linkage dimensions on gait characteristics such as stride length and gait height.

The project demonstrated the effectiveness of single degree-of-freedom linkage systems for generating walking trajectories with reduced control complexity.

Chapter 13

Future Scope

Possible future improvements include:

- Dynamic force analysis
- Real-time hardware implementation
- Human gait optimization
- Genetic algorithm based parameter tuning
- Three-dimensional mechanism analysis
- Robotic walking platform integration

Chapter 14

Member Contributions

- **Hari Patel** contributed to MATLAB simulation development, mechanism analysis, debugging, and report preparation.
- **Yug Gabani** contributed to kinematic modeling, trajectory analysis, and parameter testing.
- **Kartik Shah** contributed to mechanism visualization, simulation verification, and result analysis.
- **Prince Savaliya** contributed to simulation testing, trajectory comparison, and documentation support.
- **Jay Vaghela** contributed to literature study, experimental observations, and validation of simulation outputs.

Chapter 15

References

1. Theo Jansen mechanism and walking linkage literature.
2. MATLAB Documentation: <https://www.mathworks.com/>
3. Mechanism and Machine Theory references for linkage kinematics.
4. Project files, MATLAB codes, and simulation outputs available at: [YOUR_GOOGLE_DRIVE_LINK](#)